Question	Answer	Marks
1	**Defining the problem**	
	f is the independent variable and Q is the dependent variable, or vary f and measure Q.	**1**
	Keep h constant	**1**
	Methods of data collection	
	Labelled diagram of workable experiment including: • fan is positioned in line with the turbine so that blades of both fan and turbine overlap • base of fan on same bench as turbine • fan labelled and one other label from bench, (wind) turbine, cable, pump, pipe, liquid	**1**
	Labelled apparatus showing workable method to collect the liquid from the top of the pipe, e.g. hose / pipe / tube connected to top of pipe with the other end over a beaker / measuring cylinder below top of pipe. At least one label related to collection of liquid.	**1**
	Use of stop-watch / timer to measure time to collect liquid **or** to measure time for blades to rotate.	**1**
	Use of (top pan) balance to measure mass of liquid leaving the pipe.	**1**
	Method of Analysis	
	Plots a graph of Q against f^3 or equivalent (e.g. f^3 against Q). Do not accept logarithmic graphs.	**1**
	$C = gh \times y$-intercept (for f^3 against Q: $C = -D \times y$-intercept)	**1**
	$D = gh \times$ gradient (for f^3 against Q: $D = \frac{gh}{\text{gradient}}$)	**1**

https://xtremepape.rs/

Question	Answer	Marks
1	**Additional detail including safety considerations** Any **six** from:	**6**
	D1 Precaution with reason linked to prevent liquid spilling (on bench / floor) e.g. use of large bucket / bowl / tray to contain any spilled liquid **or** Precaution with reason linked to prevent air / dust particles in eye, e.g. use of goggles **or** Precaution with reason linked to turbine falling, e.g. clamp turbine to bench.	
	D2 Use rule to measure *h*.	
	D3 Method to determine mass of liquid, e.g. mass of beaker + liquid – mass of empty beaker **or** mass of container / pipe before – mass of container / pipe after.	
	D4 Method to determine *f*, e.g. measure time *t* for many rotations / revolutions *N* and period $T = t/N$ and $f = 1/T$ **or** measure time *t* for many rotations / revolutions *N* and $f = N/t$ **or** video rotating blades, playback frame by frame and use a time stamp to determine period *T* and $f = 1/T$.	
	D5 Mark one of the blades to assist in counting number of rotations.	
	D6 Method to vary *f*, e.g. change speed of fan / change distance between fan and blades / vary current in fan.	
	D7 Wait for steady air flow before starting timing and / or collecting liquid.	
	D8 $Q = \frac{\text{mass (of liquid)}}{\text{time (to collect liquid)}}$ **or** method and explanation to reduce uncertainty in *Q*, e.g. use large value of time or mass of liquid collected.	
	D9 Repeat measurements of *Q* for the same value of *f* **and** average *Q*.	

https://xtremepape.rs/

Question	Answer	Marks
1	D10 Relationship valid <u>if</u> a straight line is produced (not passing through the origin). Do not accept passing through the origin.	

Question	Answer	Marks
2(a)	gradient = $\frac{v}{4}$ y-intercept = $-k$	**1**
2(b)	<table><tr><th>$1/f$ / 10^{-3} Hz^{-1}</th><th>d / cm</th></tr><tr><td>0.67 or 0.667</td><td>24.7 ± 0.2</td></tr><tr><td>0.48 or 0.476</td><td>17.4 ± 0.2</td></tr><tr><td>0.36 or 0.357</td><td>12.7 ± 0.3</td></tr><tr><td>0.24 or 0.244</td><td>8.4 ± 0.3</td></tr><tr><td>0.19 or 0.192</td><td>6.6 ± 0.4</td></tr><tr><td>0.13 or 0.132</td><td>4.6 ± 0.4</td></tr></table> First mark: values of $1/f$ and d correct as shown. Second mark: uncertainties in d correct as shown.	**2**
2(c)(i)	Six points from **(b)** plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	**1**
	Error bars in d / cm plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	**1**

https://xtremepape.rs/